

SAGE: Ergodic Control for Autonomous and Adaptive Inspection of Subsea Infrastructure

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Abstract—Subsea Christmas Trees (XTs) are underwater structures that use valves for directing oil flow, needing constant inspection. But not every valve carries the same risk at the same time: a valve with a suspected leak needs to be revisited far more often than one with a clean history, and that risk picture changes during the mission as new leaks are found. To handle this, we present SAGE (Semantic and Adaptive Generative Ergodicity), an ergodic-control architecture that allocates vehicle time in proportion to a live, sensor-derived risk distribution rather than a scripted route. We study a two-XT scenario, with five valves in total, and compare a fixed-loop A* tour against SAGE. Both methods can be tuned to spend similar total time near a high-risk valve, but only ergodic control also checks it more often: in simulation, a dominant-risk valve was revisited every 5.8 s under ergodic control against a fixed 8.1 s for every valve under A*, regardless of risk, so a leak can go unnoticed for barely two-thirds as long. Because the tracked distribution is recomputed rather than planned once, a newly detected leak shifts vehicle behavior on the next control cycle with no explicit re-planning step and no operator in the loop, which a fixed tour cannot do without a discrete re-route. We derive the ergodic control law behind this behavior and report simulation results on the five-valve scenario.

I. INTRODUCTION

Inspecting subsea production infrastructure means repeatedly checking a small set of known-risk points, valves, flanges, and connectors, rather than searching an unknown space. Today this is carried out largely by divers and remotely operated vehicles (ROVs) under direct human supervision [1], following fixed schedules rather than responding to risk as it emerges. The difficulty is dividing attention among them when their relative risk is uneven and non-stationary: a leak detected mid-mission should immediately pull more of the vehicle’s time, without a human re-planning the route. A fixed inspection loop treats every point the same, and making it risk-aware normally means adding a rule that turns severity into a wait time at each point, then redoing the route by hand whenever risk changes. Ergodic control addresses this problem directly. Instead of planning a route, it acts as an **autonomous** coverage controller. It aligns the time-averaged statistics of the trajectory with a target risk distribution. This means that revisit frequency, not just time spent, measures importance automatically and adjusts within a control cycle. **No human is involved in this process.**

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II. SCENARIO AND ERGODIC CONTROL

Fig. 1 shows the target scenario: two subsea XTs, one with two monitored valves and the other with three, five points of interest in total, non-collinear and unevenly spaced. Each valve i carries a time-varying risk weight $w_i(t)$; in the full SAGE architecture (Fig. 1, right) this weight updates autonomously from operator intent, from a camera or sonar flagging a new leak, or from both fused together. For the benchmark below it is instead hand-set to fixed values per scenario, so the coverage behaviour can be validated against a known risk profile before closing the perception loop.

Unlike a path planner, the controller never represents a route: it turns the current coverage error into a velocity every cycle, so the trajectory emerges rather than being planned. Let $X \subset \mathbb{R}^2$ be the normalized workspace and $\phi(x, t)$ the current risk over that workspace, built from a weighted mixture of the five valve positions,

$$\phi(x, t) = \frac{1}{Z} \sum_i w_i(t) \mathcal{N}(x; \mu_i, \Sigma_i), \quad (1)$$

with $\mathcal{N}(x; \mu_i, \Sigma_i)$ a Gaussian density, so valve i contributes a bump at its known position μ_i , of spread Σ_i , scaled by its live risk weight $w_i(t)$; Z rescales the sum so ϕ integrates to one. A trajectory is ergodic with respect to ϕ when the fraction of time it spends in any region converges to the risk mass ϕ assigns there [3], so a valve carrying twice the risk should, over time, receive roughly twice the visits.

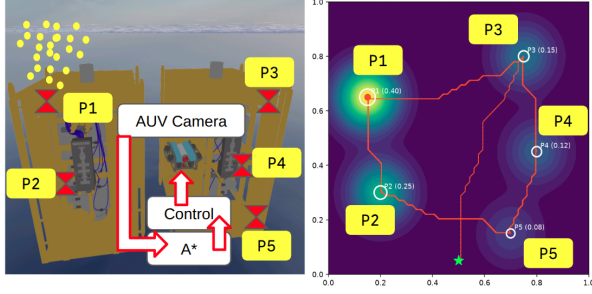
Comparing a distribution against a trajectory directly is awkward, so both are projected onto the same Fourier basis $F_k(x) = \frac{1}{h_k} \prod_j \cos(k_j \pi x_j / L_j)$: $\phi_k = \int_X \phi(x) F_k(x) dx$ gives the target’s spectral coefficients, and $c_k(t) = \frac{1}{t} \int_0^t F_k(x(\tau)) d\tau$ gives the same for the trajectory’s own running time-average. Their weighted mismatch, emphasizing the coarse shape of ϕ over its fine detail, is exactly the ergodic metric of Mathew and Mezić [3]. Rather than minimizing that metric over the whole mission, the spectral multiscale coverage law of Mathew and Mezić [3], which we implement on top of the reference ergodic-control-sandbox [4], drives it down one instant at a time: the vehicle moves along $S(x, t)$ at constant transit speed, where

$$S(x, t) = - \sum_k \Lambda_k (c_k(t) - \phi_k) \nabla_x F_k(x), \quad (2)$$

$$\Lambda_k = (1 + \|k\|^2)^{-\frac{n+1}{2}},$$

downweights high frequencies so coarse coverage is corrected before fine detail. Because ϕ enters S only through ϕ_k , raising

Baseline: Planned-Route Loop



SAGE (Ours): Autonomous Coverage Loop

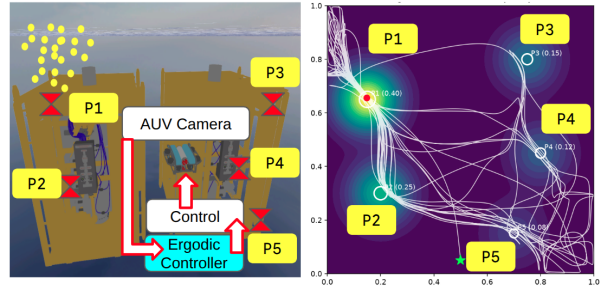


Fig. 1: Valve P1 leak response during BlueROV2 inspection of two subsea XTs in Gazebo [2]. **Left (A*)**: The leak spikes risk at P1, but adapting requires stopping to manually re-plan and re-issue routes; all valves remain at a fixed 8.1 s visit interval. **Right (SAGE, ours)**: The risk field updates dynamically without re-planning, automatically re-allocating coverage on the next cycle to revisit P1 every 5.8 s (about $1.4\times$ more often than A*’s fixed 8.1 s) fully autonomously.

TABLE I: One-extreme profile (leak at valve P1), per-valve outcome, A* vs. ergodic control (EC), 300 s mission.

Valve	Weight	A*		EC	
		visits	interval	visits	interval
P1 (leak)	0.55	37	8.1 s	50	5.8 s
P2	0.20	37	8.1 s	10	26.8 s
P3	0.10	37	8.1 s	0	—
P4	0.08	37	8.1 s	7	43.5 s
P5	0.07	37	8.1 s	15	18.4 s

any $w_i(t)$ changes the commanded direction on the very next control cycle. This is what lets a newly flagged leak redirect the vehicle immediately: reflecting the same event in an A* tour instead means detecting it, recomputing wait times, and re-issuing a route, a discrete step that normally waits on an operator or supervisory trigger, whereas here coverage, prioritization, and reprioritization all live inside one continuous control law with no planning step anywhere.

III. RESULTS AND DISCUSSION

The baseline is a single closed-loop A* tour visiting all five valves once per lap on an 8-connected grid (shortest-tour order), with per-valve wait time set proportional to w_i so that A* can match ergodic control on time spent per valve; its revisit interval, however, is fixed by lap geometry at 8.1 s regardless of weight, since the loop itself encodes no priority. We ran the five-valve layout of Fig. 1 for 300 s under three hand-set risk profiles: equal weights (20% each, a sanity check), graded weights (40/25/15/12/8%), and the one-extreme profile of Table I (55/20/10/8/7%), representing a single confirmed leak at valve P1 against four routine valves.

Table I presents the results. A* revisits every valve 37 times over the mission regardless of risk, while ergodic control tracks the same weights, checking leaking valve P1 50 times (every 5.8 s) against zero visits at P3. Correlating revisit count with weight is undefined for A* (zero variance across valves) but $+0.88$ (graded) and $+0.93$ (one-extreme) for ergodic control. P3’s zero visits are a real limitation. Unconstrained ergodic control only bounds the long-run average revisit rate, not any single point’s rate within a finite mission, so a strongly

peaked risk distribution can starve the lowest-weight valves. The realized trajectories are shown in Fig. 1 for the graded-weight case; the brief corner excursions visible in the ergodic panel are a small Fourier-basis artifact (2–5% of mission time), not a bug, and do not affect the numbers in Table I.

The lack of visits to P3 in Table I is a real limitation. One way to solve this would be to inc a minimum weight floor ($w_i(t) \geq \epsilon > 0$) or a hard revisit-interval constraint layered on the ergodic objective would guarantee every valve is checked at least once per mission, and is a natural next addition to the control law. Beyond that, the next steps are to test a broader set of scenarios and XT layouts, close the perception loop by driving $w_i(t)$ directly from real-time camera-based leak detection instead of the hand-set profiles used here, and validate all of this in real-world trials on a physical vehicle, including a mid-mission leak-injection experiment.

A more complete version of SAGE would integrate detection confidence into $w_i(t)$ using a Bayesian update and determine operator intent with a language model, as shown in Fig. 1 (right). Since the controller only requires a single risk weight per point of interest rather than specific geometry for a task, the same structure can apply beyond valve inspection. It can be used for other coverage tasks determined by a risk or interest field, such as search-and-rescue casualty localization, monitoring environmental hazards like harmful algal blooms or oil spills, or litter collection by an autonomous surface vehicle (ASV) in fjords and coastal waters.

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